

Signal Denoising

a signal $x \in \mathbb{R}^d \rightarrow$

$$y = x + \varepsilon \begin{matrix} \xrightarrow{\text{noise}} \\ \searrow \text{noisy observation.} \end{matrix}$$

①

Goal: reconstruct x .

Ψ : an orthonormal transformation

$$\theta = \Psi^T x \in \mathbb{R}^d \quad \text{supp}(\theta) \ll d$$

$$\Rightarrow \tilde{y} = \theta + \tilde{\varepsilon} \Leftarrow$$

If θ^* is known to be sparse. $\Rightarrow$ Thresholding estimators

hard-thresholded : $H_{\lambda}(y)_i = \begin{cases} y_i & \text{if } |y_i| \geq \lambda \end{cases} \leftarrow$

soft-thresholded : $S_{\lambda}(y)_i = \begin{cases} \text{sign}(y_i)(y_i - \lambda) & \text{if } |y_i| \geq \lambda \\ 0 & \text{o.w.} \end{cases} \leftarrow$

H and S are both estimators of θ .

a)

$$\min_{\theta \in \mathbb{R}^n} \left\{ \frac{1}{2} \|\gamma - \theta\|_2^2 + \frac{1}{2} \lambda^2 \|\theta\|_0 \right\}$$

Proof:

$$\gamma = (\gamma_1, \dots, \gamma_n)$$

$$\theta = (\theta_1, \dots, \theta_n)$$

$$= \sum_{i=1}^n \underbrace{\left[\frac{1}{2} (\gamma_i - \theta_i)^2 + \frac{\lambda^2}{2} \mathbb{1}[\theta_i \neq 0] \right]}_{f_i(\theta_i)}$$

for each i :

$$\min_{\theta_i \in \mathbb{R}} f_i(\theta_i) = \min_{\theta_i \in \mathbb{R}} \left\{ \frac{1}{2} (\gamma_i - \theta_i)^2 + \frac{\lambda^2}{2} \mathbb{1}[\theta_i \neq 0] \right\}$$

Case 1: $\theta_i \neq 0$;

$$\frac{\lambda^2}{2} + \frac{1}{2} \min_{\theta_i \in \mathbb{R}} (\gamma_i - \theta_i)^2 \Rightarrow \hat{\theta}_i = \gamma_i \quad (\text{where } \gamma_i \neq 0)$$

$$\rightarrow f_i(\theta_i) = \frac{\lambda^2}{2}$$

Case 2: $\theta_i = 0$

$$\Rightarrow f_i(\theta_i) = \frac{1}{2} \gamma_i^2$$

$$\text{If } \frac{\lambda^2}{2} \leq \frac{\gamma_i^2}{2} \Rightarrow \hat{\theta}_i = \gamma_i \quad \text{o.w. it's } \hat{\theta}_i = 0$$

b)

$$\min_{\theta \in \mathbb{R}^n} \left\{ \frac{1}{2} \|\gamma - \theta\|_2^2 + \lambda \|\theta\|_1 \right\}$$

Proof:

$$\hookrightarrow = \sum_{i=1}^n \left[\overbrace{\frac{1}{2} (y_i - \theta_i)^2}^{\text{convex}} + \overbrace{\lambda |\theta_i|}^{\text{convex}} \right]$$

$$\min_{\theta_i \in \mathbb{R}} \left\{ g_i(\theta_i) \right\}$$

Subgradient.

$f: \mathbb{R}^n \rightarrow \mathbb{R}$ convex.

a vector $g \in \mathbb{R}^n$ is a subgradient of f at x , if $\forall z$:

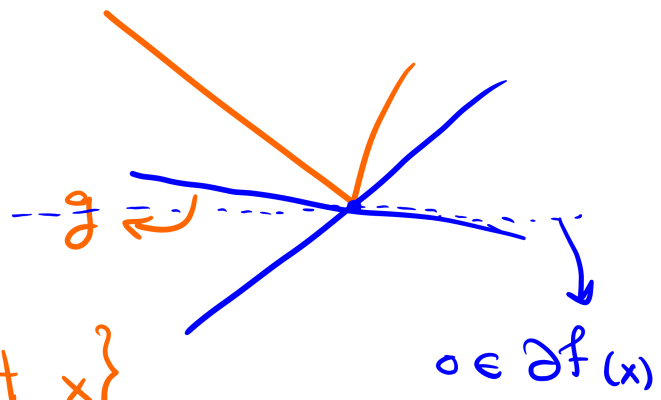
$$f(z) \geq f(x) + g^T(z - x)$$

subdifferential.

$$\partial f(x) = \{ g \mid g \text{ is subgradient of } f \text{ at } x \}$$

if differentiable.

$$\partial f(x) = \{ \nabla f(x) \}$$



A ^{Point} x^* is the global optimizer of $f \iff 0 \in \partial f(x^*)$.

$$0 \in \partial g_i(\theta_i) = (\theta_i - \gamma_i) + \lambda \partial |\theta_i|$$

$$\partial |\theta_i| = \begin{cases} \{-1\} & \theta_i < 0 \\ [-1, 1] & \theta_i = 0 \\ \{1\} & \theta_i > 0 \end{cases}$$

Case 1: $\theta_i > 0$: $0 = \theta_i - \gamma_i + \lambda \Rightarrow \theta_i = \gamma_i - \lambda$

$\gamma_i > \lambda$

Case 2: $\theta_i < 0$: $\Rightarrow \theta_i = \gamma_i + \lambda \rightarrow \gamma_i < -\lambda$

Case 3: $\theta_i = 0$: $\Rightarrow 0 \in \gamma_i + \lambda[-1, 1] \Rightarrow \gamma_i \in [-\lambda, \lambda]$
 $\Rightarrow |\gamma_i| \leq \lambda$

$$\hat{\Theta}_i = \begin{cases} \text{sign}(\gamma_i) (|\gamma_i| - \lambda) & |\gamma_i| \geq \lambda \\ 0 & \text{o.w.} \end{cases}$$

It's makes sense to assume θ^* is sparse & satisfies approximate sparsity.

Bound for s-term approximation. approx. support of θ^*

$$s \in \{1, \dots, d\}$$

$$S \subseteq \{1, \dots, d\}$$

$$\begin{array}{c} \uparrow \\ |S| \leq s \end{array}$$

$$\theta^* \in \mathbb{R}^d$$

$$\Pi_s(\theta^*) = \underset{\|\theta\|_0 \leq s}{\operatorname{argmin}} \|\theta - \theta^*\|_2^2$$

Closed form : Fix $S \subseteq \{1, \dots, d\}$ s.t. $|S| \leq s$

$$\hat{\theta}_i = \begin{cases} \theta_i^* & i \in S \\ 0 & i \notin S \end{cases}$$

error = $\sum_{j \notin S} |\theta_j^*|^2$

we should choose S to maximize $\sum_{j \in S} |\theta_j^*|^2 \Rightarrow S_s(\theta^*)$

$$\pi_s(\theta^*)_j = \begin{cases} \theta_j^* & j \in S_s(\theta^*) \\ 0 & \text{o.w.} \end{cases}$$

$$B_q(R_q) = \left\{ \theta \in \mathbb{R}^d \mid \sum_{j=1}^d |\theta_j|^q \leq R_q \right\} \leftarrow$$

$$\underbrace{|\theta^*|_{(1)}}_{\downarrow \max_j |\theta_j^*|} \geq \dots \geq |\theta^*|_{(d)} \xrightarrow{\downarrow} \min_j |\theta_j^*| \rightarrow \text{our support.}$$

$$|\theta^*|_{(1)}, \dots, |\theta^*|_{(s)}$$

$$\|\Pi_s(\theta^*) - \theta^*\|_2^2 = \sum_{j=s+1}^d |\theta^*|_{(j)}^2$$

$$a = |\theta^*|_{(s)} \geq |\theta^*|_{(j)} \quad \text{for } j > s.$$

$$a = |\theta^*|_{(s)} \leq |\theta^*|_{(j)} \quad \text{for } j \leq s.$$

$$s a^q \leq \sum_{j=1}^s |\theta^*|_{(j)}^q \leq \sum_{j=1}^d |\theta^*|_{(j)}^q \leq R_q$$

$$\Rightarrow a^q \leq \frac{R_q}{s} \quad (*)$$

With assumption that $\theta^* \in \mathcal{B}_q(R_q)$ we can write:

$$\|\Pi_s(\theta^*) - \theta^*\|_2^2 = \sum_{j=s+1}^d |\theta^*|_{(j)}^2$$

$$|\theta^*|_{(i)} \leq |\theta^*|_{(s)} = a \quad \leftarrow \quad = \sum_{j=s+1}^d |\theta^*|_{(j)}^q \underbrace{|\theta^*|_{(j)}^{2-q}}$$

$$\leq \sum_{j=s+1}^d a^{2-q} |\theta^*|_{(j)}^q$$

$$\leq a^{2-q} R_q$$

$$\begin{aligned}
 & \stackrel{(*)}{\implies} \| \pi_s(\theta^*) - \theta^* \|_2^2 \leq a^{2-q} R_q = (a^q)^{\frac{2}{q}-1} R_q \\
 & \stackrel{(*)}{\leq} \left(\frac{R_q}{s} \right)^{\frac{2}{q}-1} R_q \\
 & = R_q^{2/q} \left(\frac{1}{s} \right)^{\frac{2}{q}-1}
 \end{aligned}$$